SMART WEB SECURITY DETECTION SYSTEM

Shree Priya - 2303A51465
Anushka - 2303A510E6
Sushwanth - 2303A51567
Anvika - 2303A510C9

INTRODUCTION

- Phishing is a common cyber attack where attackers create fake websites that look identical to real ones.
- These websites trick users into entering sensitive information such as passwords, banking details, and personal data.
- With the increase in online activities, phishing attacks are becoming more frequent and dangerous.
- CyberShield is designed to detect such phishing websites by analyzing URL patterns and providing immediate results to users.

OBJECTIVES

The main objectives of this project are:

- To identify phishing websites using rule-based URL analysis
- To provide instant feedback (Safe / Phishing)
- To display a risk score for better understanding
- To store scanned URLs for future reference
- To create a simple and user-friendly web interface
- To improve awareness about cyber security among users

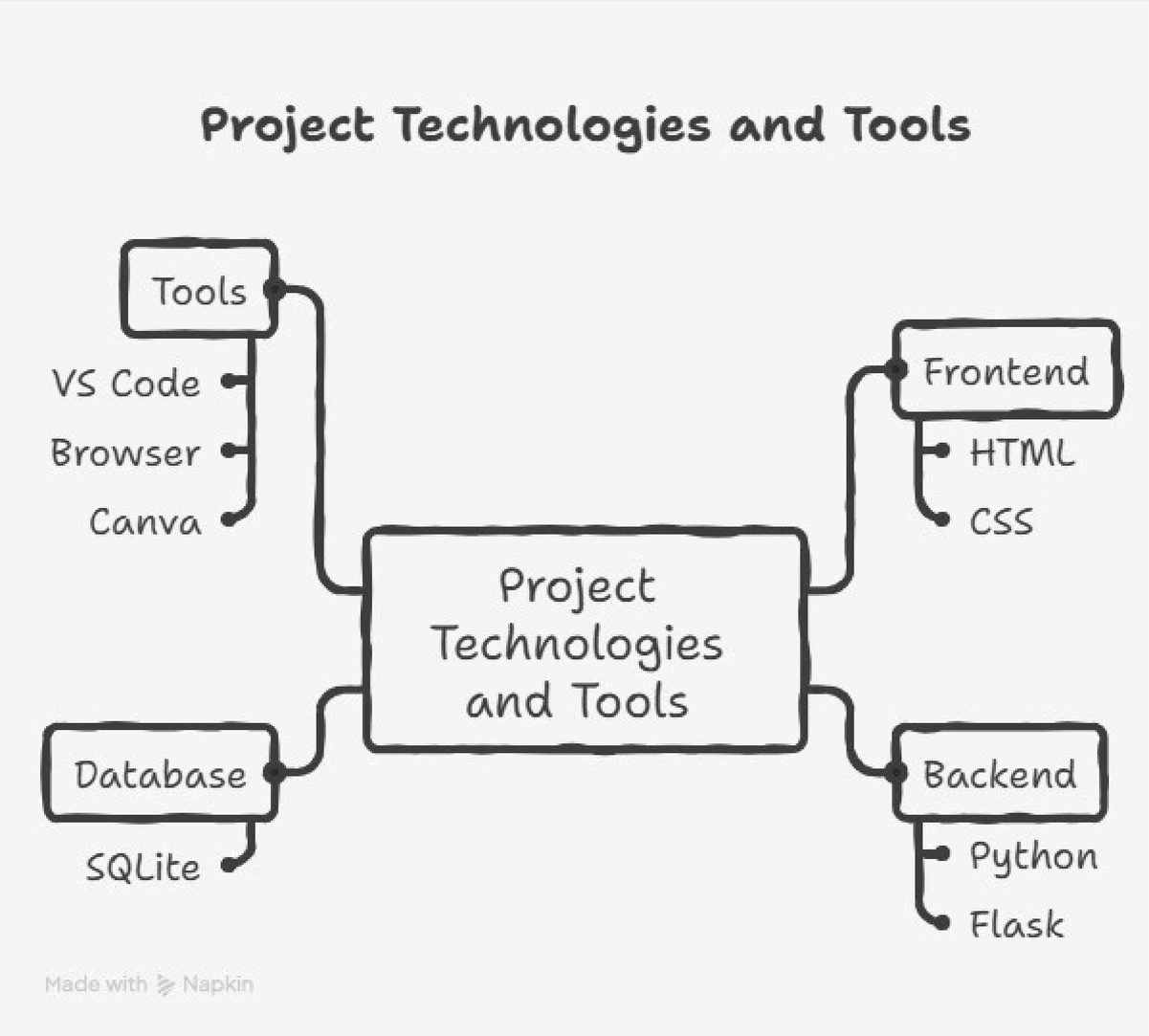
STAKEHOLDERS

The stakeholders of this project include:

- Internet Users – for safe browsing experience
- Students – to learn about phishing and cyber threats
- Cyber Security Professionals – as a basic analysis tool
- Organizations – to protect employees and users from phishing attacks
- Developers – to understand web security concepts

TOOLS & TECHNOLOGIES

- Frontend: HTML (structure), CSS (design)
- Backend: Python, Flask (handles requests)
- Database: SQLite (stores URL results)
- Tools: VS Code, Browser, Canva



AI PROMPTS USED

AI tools like ChatGPT were used during development:

Some prompts used:

- "Create a phishing detection system using Python Flask"
- "Design a clean and modern UI using HTML and CSS"
- "Add risk score and URL validation logic"
- "Improve user interface and add history feature"

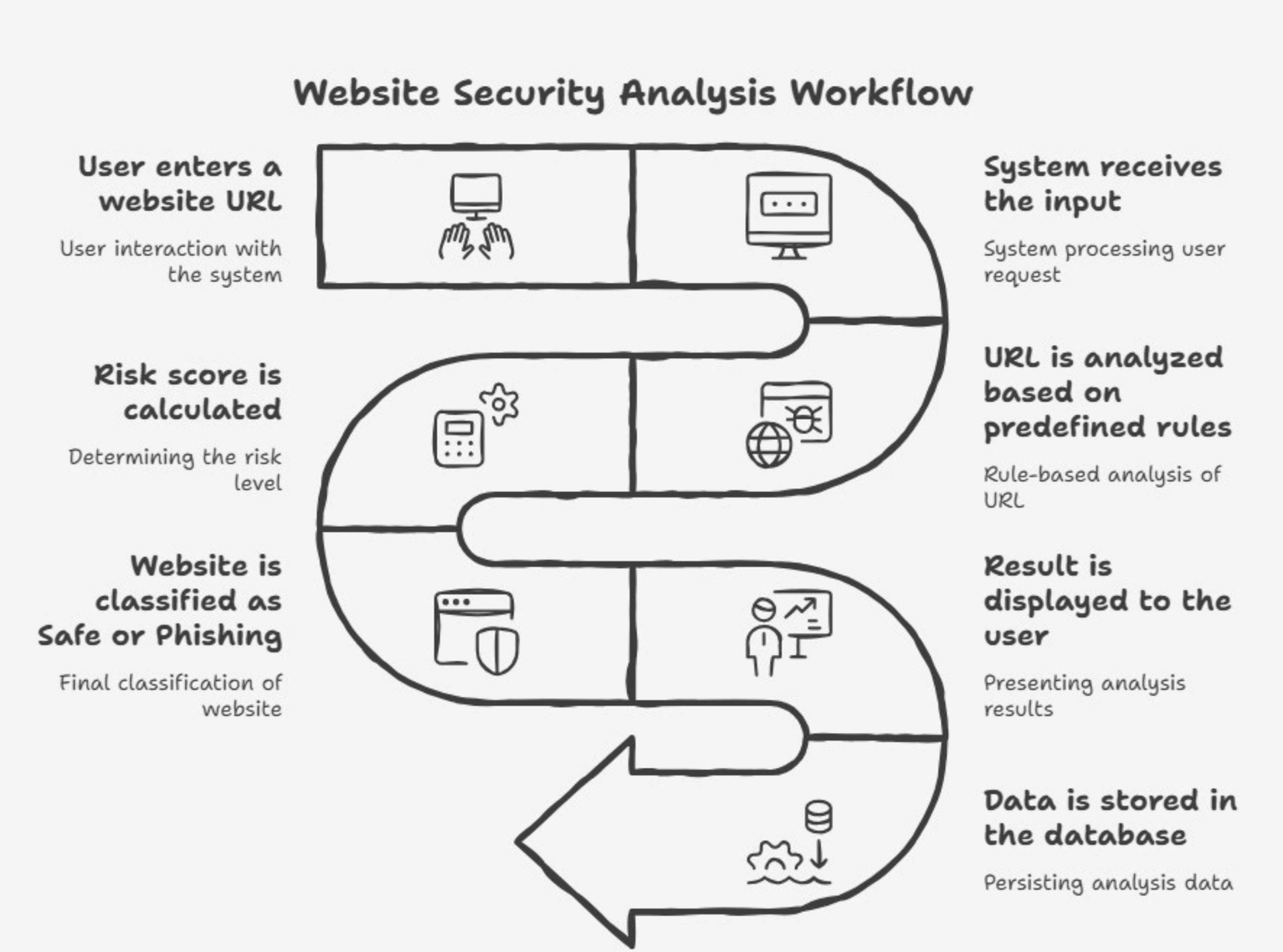
AI helped in:

- Faster code generation
- Improving UI design
- Debugging errors
- Enhancing overall project quality

FLOWCHART

Workflow of the system:

1. User enters a website URL
2. System receives the input
3. URL is analyzed based on predefined rules
4. Risk score is calculated
5. Website is classified as Safe or Phishing
6. Result is displayed to the user
7. Data is stored in the database



FRONTEND VIEW

Features of the frontend:

- Input field to enter website URL
- Scan button to analyze the URL
- Result display (Safe / Phishing)
- Risk score visualization
- Clean and responsive design
- Navigation bar for easy access
- History page to view previous scans

PROOF OF PROMPTS

🧠 BACKEND (Flask - Python)

Python

```python
def check_phishing(url):
    score = 0

    if "@" in url:
        score += 3
    if "-" in url:
        score += 1
    if "login" in url or "verify" in url:
        score += 2
    if url.startswith("http://"):
        score += 2
    if not url.startswith("http"):
        score += 2

    if score > 5:
        return "⚠️ Phishing Website", "danger", score
    else:
        return "✅ Safe Website", "safe", score
```

🌐 FRONTEND (HTML)

HTML

```html
{% if result %}
<div class="result {{ status }}">
{{ result }}
</div>

<p>Risk Score: {{ score }}/10</p>
{% endif %}
```

- Build phishing detection system using Flask
- Add risk score logic
- Design frontend using HTML & CSS
- Store data using SQLite
- Improve UI and user experience

REFERENCES

- Python Official Documentation
  https://docs.python.org/
- Flask Official Website
  https://flask.palletsprojects.com/
- W3Schools (HTML, CSS, JavaScript)
  https://www.w3schools.com/
- ChatGPT (AI Tool for Code Generation)
  https://chat.openai.com/
- Visual Studio Code Documentation
  https://code.visualstudio.com/docs
- Online Cyber Security Articles & Tutorials
  (Google, Blogs, Learning Platforms)